Finding the Nuclei That Models Miss

Reliable instance segmentation and classification
across diverse tissues in PanNuke

Biomedical Data Design I | EN.580.437/697 | Fall 2026

Project 5

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Project proposal | October 14, 2026

Project summary

Automated tissue analysis depends on accurate counts and types of individual nuclei. Yet strong average segmentation scores can conceal substantial under-detection of rare, small, or partially visible cells. In our PanNuke experiments, this deficit is particularly pronounced for the Dead class, which includes apoptotic and necrotic nuclei.

We propose a study of rare-nucleus detection that connects evidence from reproduced baselines to two complementary interventions: supervision for cropped nuclei and direct instance prediction. The supervision study tests how training targets affect partially visible objects. PM-Net, our proposed architecture, instead uses point-conditioned queries to predict individual masks and cell types, replacing horizontal/vertical distance maps and watershed decoding. Matched ablations will establish whether the new decoder improves detection without sacrificing boundary quality or imposing excessive inference cost.

Intellectual merit

The central question is how instance representation affects the detection of rare and partially visible nuclei. By comparing a targeted supervision change with a direct mask decoder under the same encoder and evaluation protocol, the study will distinguish gains due to target construction from those due to model architecture.

Broader impacts

More reliable nucleus-level measurements can improve studies of tissue composition and the tumor microenvironment. Shared evaluation code and tissue-specific error analyses will support reproducible comparisons, with rare-class detection assessed alongside segmentation quality and computational cost.

1. Research aims, context, and preliminary evidence

Hematoxylin and eosin (H&E) images contain adjacent nuclei that vary in staining, morphology, and tissue context. Missed apoptotic or necrotic nuclei, merged neighbors, and type confusion can bias estimates of cell composition. PanNuke provides instance masks and five cell labels across 19 tissues, making it a useful setting for studying these errors [1].

Specific aims

Aim 1 - Characterize segmentation and typing failures. Consolidate HoVer-Net and CellViT-UNI results under a shared evaluator, separating missed nuclei, merges, splits, and type errors by tissue and class.

Aim 2 - Evaluate supervision for truncated nuclei. Complete the four-arm, two-seed comparison of stronger cropping and selective HV supervision, using both fixed crop windows and original validation images.

Aim 3 - Develop a point-conditioned instance model. Test whether PM-Net improves Dead detection and crowded-instance separation within predefined quality and inference-cost limits.

Prior work and architectural motivation

HoVer-Net separates instances using horizontal/vertical (HV) offset maps [2], while CellViT combines a transformer encoder with nuclei segmentation and classification [3]. Our reproduced CellViT-UNI is the primary comparison model. PM-Net draws on point-conditioned nucleus masks in PromptNucSeg [4], query-based mask prediction in Mask2Former [5], and spatial adapters in CellVTA [6]. It combines these ideas around a shared UNI encoder so that the initial study isolates decoder design.

Data and baseline evidence

PanNuke contains 7,901 RGB patches at 256 x 256 pixels and roughly 200,000 annotated nuclei. Its three prepared folds contain 2,656, 2,523, and 2,722 patches. Each instance is labeled neoplastic, inflammatory, connective, dead, or epithelial. Development uses fold 1 for training and fold 2 for validation.

Reproduced baselines: three-split means	mPQ	bPQ
HoVer-Net, final checkpoint, no test-time augmentation	0.4564	0.6613
CellViT-UNI, final checkpoint, no test-time augmentation	0.4995	0.6653

Earlier analysis found that 38.5% of labeled Dead nuclei had intersection-over-union (IoU) below 0.1 with every predicted object. Reclassification, copy-paste augmentation, synthesis, and frozen-feature probes did not yield reliable improvements under the tested recipes [7]. Two-scale prediction produced a small Dead PQ gain at about 6.1 times the inference cost. These findings focus the present study on instance detection and partial visibility rather than further type recalibration.

2. Aim 2: supervision for truncated nuclei

Spatial cropping truncates nuclei at training-image boundaries. Under the existing target rule, fragments below 30 pixels receive no HV separation target. Aim 2 tests whether retaining that target for newly truncated fragments improves detection, with identical images and other labels across the matched intervention arms.

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective supervision	0.5	Minimum 1 for newly truncated instances with visible area below 30
A3: small-object control	0.5	Minimum 1 for all instances; targets differ only below area 30

All arms use seeds 19 and 1, 130 epochs, matched initialization and optimization, and final-checkpoint evaluation at native resolution without test-time augmentation (TTA). A1-A3 share the augmentation random sequence. Instance identities before and after cropping distinguish newly truncated objects from those already cut by a patch boundary; areas are measured after geometric transformations [8].

Interim findings

The October 14 analysis includes six completed fold-2 evaluations: all four arms for seed 19, and A0/A1 for seed 1. Relative to A0, A2 seed 19 raises Dead PQ by 0.0222 and strict Dead PQ by 0.0025, but lowers mPQ by 0.0091 and bPQ by 0.0062. These losses exceed the predefined 0.002 quality margin, so this variant does not qualify for further expansion under the existing protocol [8].

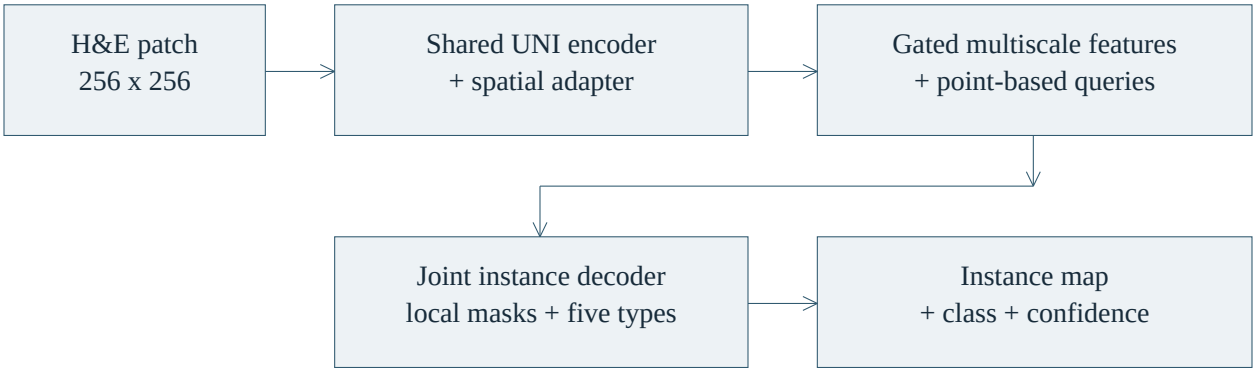
Mechanism analysis

The remaining analysis uses 2,523 fixed crop windows to measure misses among newly truncated Dead nuclei. A2-A1 isolates selective supervision, A2-A0 quantifies its overall tradeoff, and A3 tests whether the effect extends to all small instances. The frozen rule requires fewer misses in both seeds, a mean reduction of at least 0.03, and a positive 95% lower confidence bound. Additional Dead and strict-score conditions remain as specified in the original protocol [8].

The observed tradeoff raises a broader design question: can a model represent a partially visible instance without relying on a dense geometric separation field? Aim 3 examines this question through direct mask prediction. Aim 2 provides a controlled supervision comparison; PM-Net tests a distinct representation hypothesis, with the encoder held constant.

3. Aim 3: PM-Net, a direct instance model

PM-Net will predict an individual mask and cell type from each candidate nucleus location [9]. A shared encoder supplies image context, while a spatial adapter preserves detail for small and crowded objects. The prototype will use native 256 x 256 patches and the same pretrained UNI initialization as CellViT-UNI.



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Architecture and optimization

Shared image features. A compact CNN adapter supplements UNI features with higher-resolution spatial information. Learned gates combine the feature scales before the point and mask heads.

Point-conditioned queries. A supervised heatmap predicts candidate nucleus locations and types. Each selected point initializes an instance query, a feature vector for one proposed nucleus. Training targets lie inside visible instance masks. Query capacity, initially about 256, will be set from training-set object counts with a safety margin; overflow will be reported.

Joint mask and type prediction. Queries attend to local features and tissue context to predict masks, confidence, and five-class probabilities. One-to-one matching pairs predictions with labeled nuclei using location, class, and mask agreement. Optimization combines point focal loss, mask cross-entropy and Dice losses, and type cross-entropy. Direct supervision of the point head avoids relying on gradients through discrete point selection; loss weights will be selected on validation data.

Instance decoding. A validation-selected confidence rule resolves overlapping masks, which are mapped to native image coordinates for scoring. Area filtering will be audited for its effect on small-instance recall, separately from the training-target cutoff in Aim 2.

Controlled ablations

A basic query decoder will be compared with versions that add point initialization and gated fusion separately. A plain adapter with matched parameter count will distinguish architectural effects from added capacity. The initial study retains five-class prediction throughout; truncation supervision and auxiliary type grouping are subsequent extensions rather than prerequisites for the core model.

4. Experimental design and evaluation

Development and confirmation

Development pairs seeds 19 and 1 on training fold 1 and validation fold 2, using the existing 130-epoch schedule and encoder unfreezing at epoch 25. The encoder initialization and training recipe will be matched across PM-Net and CellViT-UNI. Architecture, losses, augmentation, decoding, and decision criteria will be fixed before confirmation.

Confirmation will follow the project train/validation/test protocol: 1/2/3, 2/1/3, and 3/2/1. Final checkpoints will be evaluated for every split and seed before averaging. Splits 1 and 2 share test fold 3. Earlier stages have examined the test folds, and patient/slide identifiers are insufficient to verify patient-level separation; both limitations will accompany the results. CoNIC and MoNuSAC will provide secondary external comparisons after class mapping and overlap exclusion.

Endpoints and statistical analysis

The shared PanNuke evaluator measures official mPQ, which includes type, bPQ, which ignores type, and per-class panoptic quality (PQ). Strict mPQ and strict Dead PQ additionally retain penalties for predictions of classes absent from an image. Detection and typing errors will be separated, with Dead recall stratified by tissue and by interior versus border location. Merge/split rates and tissue-by-class heatmaps will characterize crowded-instance performance.

PM-Net will be compared with the matched CellViT-UNI baseline and the crop-only control. Paired confidence intervals will be estimated by resampling images within tissues, keeping windows from the same patch together. Results will include each seed; these intervals quantify image-sampling variation conditional on the trained models. Class counts and annotation uncertainty will guide interpretation of rare-class estimates.

PM-Net development criteria

Before training, we propose fixing a primary target of at least 0.03 mean improvement in interior Dead recall, with a positive paired 95% lower confidence bound. Mean Dead PQ must improve by at least 0.005, and both recall and Dead PQ changes must be positive in each seed. Per-seed mPQ/bPQ losses are limited to 0.002; strict mPQ and strict Dead PQ must not decline, and mean merge/split rates must not increase. These criteria apply to PM-Net, independently of the frozen supervision experiment.

The inference budget is twice the matched single-scale baseline, including decoding. Repeated latency and peak-memory measurements on the same hardware and batch size will account for competing device load. Models meeting the quality and cost criteria will proceed to confirmation.

5. Milestones, feasibility, and deliverables

Schedule	Milestone
Week 1	Complete the visibility-supervision analysis and consolidate baseline/error tables.
Week 2	Finalize PM-Net design; implement and verify matching, masks, query capacity, and runtime.
Weeks 3-4	Complete the two-seed development comparison and core ablations.
Weeks 5-6	Confirm qualifying models across splits and external datasets; complete the final report.

The six-week schedule begins at proposal agreement. Baseline reproduction and the supervision study provide the empirical foundation; the main development milestone is a tested PM-Net prototype. Cross-split confirmation and external evaluation follow the prototype decision, with auxiliary supervision reserved for a later extension. If confirmation is not warranted, Weeks 5-6 will complete the ablation and failure analysis.

Feasibility and main risks

Rare labels and annotation uncertainty. Tissue-specific counts and paired estimates will accompany rare-class results. Existing labels will be preserved, and disagreements among detectors will be retained as uncertain cases for future expert review.

Localization versus mask quality. Accurate point proposals may still produce masks below the overlap threshold, especially in crowded regions. Separate analysis of point recall, mask overlap, duplicates, and merge/split errors will identify where the query decoder succeeds or loses accuracy.

Experimental scope and resources. The encoder and training recipe remain fixed during the decoder ablations. We estimate 2-4 A100 GPU-days for development and a further 4-8 for confirmation, subject to allocation. Available storage and checkpoint requirements will be checked before scheduling; resumable training will reduce losses from interruptions.

Deliverables and broader impact

The project will deliver a tested PM-Net prototype, reproduced baselines, versioned evaluation code and configurations, model/data provenance, comparison tables, and a final report with annotated failure cases. The central contribution is an empirical account of how supervision and instance representation affect rare-nucleus detection under matched conditions.

This work supports more reliable measurements of cell composition across tissues. Reporting tissue-specific errors alongside aggregate scores will help researchers assess whether apparent improvements extend to underrepresented cells and justify their computational cost.

References and project evidence

[1] Gamper, J., et al. (2020). PanNuke Dataset Extension, Insights and Baselines.
https://arxiv.org/abs/2003.10778

[2] Graham, S., et al. (2019). HoVer-Net: Simultaneous segmentation and classification of nuclei in multi-tissue histology images. Medical Image Analysis, 58, 101563.
https://doi.org/10.1016/j.media.2019.101563

[3] Hörst, F., et al. CellViT: Vision Transformers for Precise Cell Segmentation and Classification. arXiv:2306.15350.
https://arxiv.org/abs/2306.15350

[4] Shui, Z., et al. Unleashing the Power of Prompt-driven Nucleus Instance Segmentation. arXiv:2311.15939.
https://arxiv.org/abs/2311.15939

[5] Cheng, B., et al. (2022). Masked-attention Mask Transformer for Universal Image Segmentation. CVPR.
https://arxiv.org/abs/2112.01527

[6] Yang, Y., et al. (2025). CellVTA: Enhancing Vision Foundation Models for Accurate Cell Segmentation and Classification. arXiv:2504.00784.
https://arxiv.org/abs/2504.00784

[7] bdd26 project records. Research plan, closeout audit, and stage reports I-V. Repository snapshot b60c776 (October 14, 2026).

Versioned project documentation

[8] bdd26. Visibility-supervision experiment: frozen specification (October 14, 2026), training source eee3248, and six original-image fold-2 evaluation summaries available on October 14, 2026.

Frozen supervision specification

[9] bdd26. Next-stage research proposal: a new non-HoVer-Net model route. October 14, 2026. Architectural design underlying the PM-Net prototype proposed here.

Versioned new-model concept

Team contributions to this proposal

This page is separate from the main proposal.

The team will complete the statements below before submission.

TODO - COMPLETE BEFORE COURSE SUBMISSION
Add each team member's actual contribution to this proposal.

This section is intentionally left for the team to complete. Record work on the proposal itself, such as developing the research question, reviewing literature, designing experiments, preparing figures, or drafting and revising sections.

Team member	Contribution to this proposal
Zhiyu Song	**TODO: add actual contribution.**
Yuchen Ai	**TODO: add actual contribution.**
Xinhao Jin	**TODO: add actual contribution.**
Zheng Jian	**TODO: add actual contribution.**